

Operating System (OS) Notes

Page 1: Introduction to Operating System

An Operating System (OS) is system software that acts as an interface between the user and computer hardware.

Functions: Process Management, Memory Management, File Management, Device Management, Security, User Interface.

Page 2: Types of Operating Systems

Batch OS, Multiprogramming OS, Multitasking OS.

Page 3: More Types of Operating Systems

Time Sharing OS, Real-Time OS, Distributed OS.

Page 4: Process Management

Process, PCB, Process States, Scheduling.

Page 5: CPU Scheduling

FCFS, SJF and scheduling objectives.

Page 6: CPU Scheduling Algorithms

Round Robin, Priority Scheduling.

Page 7: Memory Management

Memory allocation, fragmentation, partitioning.

Page 8: Virtual Memory and Paging

Virtual memory, pages and frames.

Page 9: File System and Deadlocks

File operations and deadlock concepts.

Page 10: Security, Protection and OS Services

Authentication, authorization, encryption, OS services.